

## TD 1

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> pwd

Path
----
C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> ls

Répertoire : C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders
```

| Mode    | LastWriteTime    | Length | Name             |
|---------|------------------|--------|------------------|
| ----    | -----            | -----  | ----             |
| d-----  | 07/10/2026 18:01 |        | config           |
| d-----  | 07/10/2026 18:01 |        | data             |
| d-----  | 07/10/2026 18:01 |        | evidence         |
| d-----  | 07/10/2026 18:01 |        | labs             |
| d-----  | 07/10/2026 18:01 |        | readiness        |
| d-----  | 07/10/2026 18:01 |        | src              |
| -a----- | 07/10/2026 18:01 | 199    | .gitignore       |
| -a----- | 07/10/2026 18:01 | 3126   | CONTRIBUTING.md  |
| -a----- | 07/10/2026 18:01 | 10747  | README.md        |
| -a----- | 07/10/2026 18:01 | 135    | requirements.txt |
| -a----- | 07/10/2026 18:01 | 1508   | TEAM_TEMPLATE.md |

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> dir

Répertoire : C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders
```

| Mode    | LastWriteTime    | Length | Name             |
|---------|------------------|--------|------------------|
| ----    | -----            | -----  | ----             |
| d-----  | 07/10/2026 18:01 |        | config           |
| d-----  | 07/10/2026 18:01 |        | data             |
| d-----  | 07/10/2026 18:01 |        | evidence         |
| d-----  | 07/10/2026 18:01 |        | labs             |
| d-----  | 07/10/2026 18:01 |        | readiness        |
| d-----  | 07/10/2026 18:01 |        | src              |
| -a----- | 07/10/2026 18:01 | 199    | .gitignore       |
| -a----- | 07/10/2026 18:01 | 3126   | CONTRIBUTING.md  |
| -a----- | 07/10/2026 18:01 | 10747  | README.md        |
| -a----- | 07/10/2026 18:01 | 135    | requirements.txt |
| -a----- | 07/10/2026 18:01 | 1508   | TEAM_TEMPLATE.md |

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> cd data
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data> pwd

Path
----
C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data

PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data> dir

Répertoire : C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data
```

| Mode   | LastWriteTime    | Length | Name   |
|--------|------------------|--------|--------|
| d----- | 07/10/2026 18:01 |        | sample |

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data> cd sample
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data\sample> pwd

Path
----
C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data\sample

PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data\sample> dir

Répertoire : C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data\sample
```

| Mode   | LastWriteTime    | Length | Name                              |
|--------|------------------|--------|-----------------------------------|
| -a---- | 07/10/2026 18:01 | 7625   | bloomberg_reference_expected.json |
| -a---- | 07/10/2026 18:01 | 10281  | bloomberg_reference_sample.json   |
| -a---- | 07/10/2026 18:01 | 244    | instruments.json                  |
| -a---- | 07/10/2026 18:01 | 2266   | prices.csv                        |

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders\data\sample> cd ..\..
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> pwd

Path
----
C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

Instrument : Apple Inc. (AAPL) ; Benchmark : S&P 500 (SP500) ; Emplacement : Dans data/sample/instruments.json

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> Get-Content data\sample\prices.csv -Head 10
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> |
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> Select-String "AAPL" data\sample\prices.csv
```

```
data\sample\prices.csv:2:2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
data\sample\prices.csv:4:2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
data\sample\prices.csv:6:2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
data\sample\prices.csv:8:2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
data\sample\prices.csv:10:2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
data\sample\prices.csv:12:2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
data\sample\prices.csv:14:2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
data\sample\prices.csv:16:2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
data\sample\prices.csv:18:2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
data\sample\prices.csv:20:2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
data\sample\prices.csv:22:2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
data\sample\prices.csv:24:2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
data\sample\prices.csv:26:2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
data\sample\prices.csv:28:2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
data\sample\prices.csv:30:2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
data\sample\prices.csv:32:2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
data\sample\prices.csv:34:2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
data\sample\prices.csv:36:2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
data\sample\prices.csv:38:2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
data\sample\prices.csv:40:2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
data\sample\prices.csv:42:2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> Select-String "SP500" data\sample\prices.csv
```

```
data\sample\prices.csv:3:2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
data\sample\prices.csv:5:2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
data\sample\prices.csv:7:2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
data\sample\prices.csv:9:2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
data\sample\prices.csv:11:2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
data\sample\prices.csv:13:2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
data\sample\prices.csv:15:2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
data\sample\prices.csv:17:2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
data\sample\prices.csv:19:2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
data\sample\prices.csv:21:2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
data\sample\prices.csv:23:2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
data\sample\prices.csv:25:2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
data\sample\prices.csv:27:2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
data\sample\prices.csv:29:2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> python --version
Python 3.12.10
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> git --version
git version 2.56.0.windows.2
```

```

PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21

```

TD 2

Part 1

**Which function reads JSON?**

load\_instruments()

**Which function reads CSV?**

load\_prices()

**Which function selects one ticker?**

filter\_prices()

**Which function coordinates execution?**

main()

**Where are the input files stored?**

Dans le dossier data/sample/ (instruments.json et prices.csv).

Part 2

```

PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}

```

```
>>> import json
>>> with open("data/sample/instruments.json", encoding="utf-8") as f:
...     instruments = json.load(f)
...
>>> print(type(instruments))
<class 'dict'>
>>> print(instruments.keys())
dict_keys(['instrument', 'benchmark'])
```

```
>>> instrument = instruments["instrument"]
>>> benchmark = instruments["benchmark"]
>>>
>>> print(instrument["ticker"])
AAPL
>>> print(instrument["name"])
Apple Inc.
>>> print(instrument["currency"])
USD
>>>
>>> print(benchmark["ticker"])
SP500
>>> print(benchmark["name"])
S&P 500
```

**Qu'est-ce que instruments ?** Un dictionnaire global contenant deux objets métiers.

**Qu'est-ce que instrument ?** Un dictionnaire décrivant l'actif principal (AAPL, Apple Inc., USD).

**Qu'est-ce que benchmark ?** Un dictionnaire décrivant l'indice de comparaison (SP500, S&P 500).

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> Get-Content data\sample\p
rices.csv -Head 10
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> python
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> def load_prices():
...     with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
...         return list(csv.DictReader(file))
```

```
>>> import csv
>>> with open("data/sample/prices.csv", encoding="utf-8") as file:
...     prices = list(csv.DictReader(file))
...
>>> print(type(prices))
<class 'list'>
>>> print(type(prices[0]))
<class 'dict'>
>>> print(prices[0])
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}
```

```
>>> print(prices[0]["close"])
250.00
>>> print(type(prices[0]["close"]))
<class 'str'>
>>>
>>> close = float(prices[0]["close"])
>>> print(type(close))
<class 'float'>
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> python
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> def filter_prices(prices, ticker):
...     return [row for row in prices if row["ticker"] == ticker]
```

```
>>> import csv, json
>>>
>>> def filter_prices(prices, ticker):
...     return [row for row in prices if row["ticker"] == ticker]
...
>>> with open("data/sample/instruments.json", encoding="utf-8") as f:
...     instruments = json.load(f)
...
>>> with open("data/sample/prices.csv", encoding="utf-8") as f:
...     prices = list(csv.DictReader(f))
```

```
>>> instrument = instruments["instrument"]
>>> benchmark = instruments["benchmark"]
```

```
>>> instrument_prices = filter_prices(prices, instrument["ticker"])
>>> benchmark_prices = filter_prices(prices, benchmark["ticker"])
>>> print(len(instrument_prices))
21
>>> print(len(benchmark_prices))
21
```

```
>>> for row in prices:
...     if row["ticker"] == ticker:
...         result.append(row)
...
Traceback (most recent call last):
  File "<stdin>", line 2, in <module>
NameError: name 'ticker' is not defined
```

```

>>> def get_first_close(prices):
...     return float(prices[0]["close"])
...
>>> def get_last_close(prices):
...     return float(prices[-1]["close"])
...
>>> # Test sur AAPL
>>> print(get_first_close(instrument_prices))
250.0
>>> print(get_last_close(instrument_prices))
266.2
>>>
>>> # Test sur SP500
>>> print(get_first_close(benchmark_prices))
6600.0
>>> print(get_last_close(benchmark_prices))
6742.0

```

```
import csv
```

```
import json
```

```
from pathlib import Path
```

```
DATA_DIR = Path(__file__).resolve().parent.parent / "data" / "sample"
```

```
def load_instruments():
```

```
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
```

```
        return json.load(file)
```

```
def load_prices():
```

```
    with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
```

```
        return list(csv.DictReader(file))
```

```

def filter_prices(prices, ticker):
    return [row for row in prices if row["ticker"] == ticker]


def get_first_close(prices):
    return float(prices[0]["close"])


def get_last_close(prices):
    return float(prices[-1]["close"])


def display_market_summary(asset, prices, show_currency=True):
    ticker = asset["ticker"]
    name = asset["name"]
    count = len(prices)
    first_close = get_first_close(prices)
    last_close = get_last_close(prices)

    currency_str = f" {asset['currency']}" if show_currency and "currency" in asset else ""

    print(f"{ticker} - {name}")
    print(f"Observations : {count}")
    print(f"First close : {first_close:.2f}{currency_str}")
    print(f>Last close : {last_close:.2f}{currency_str}")


def main():

```

```
instruments = load_instruments()
```

```
prices = load_prices()
```

```
instrument = instruments["instrument"]
```

```
benchmark = instruments["benchmark"]
```

```
instrument_prices = filter_prices(prices, instrument["ticker"])
```

```
benchmark_prices = filter_prices(prices, benchmark["ticker"])
```

```
print("=== MarketPulse ===\n")
```

```
print("Market configuration")
```

```
print("Period : 1 month")
```

```
print("Interval : Daily\n")
```

```
print("Instrument")
```

```
display_market_summary(instrument, instrument_prices, show_currency=True)
```

```
print()
```

```
print("Benchmark")
```

```
display_market_summary(benchmark, benchmark_prices, show_currency=False)
```

```
if __name__ == "__main__":
```

```
    main()
```

```
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> notepad src\main.py
PS C:\Users\maely\ProjectGitLinuxPython\esilv-marketpulse-g07-Tryharders> python src\main.py
=== MarketPulse ===

Market configuration
Period   : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close  : 250.00 USD
Last close   : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close  : 6600.00
Last close   : 6742.00
```